

Answers

1.B	2.B	3.2	4.2	5.E	6.D	7.C	8.A
9.C	10.B	11.D	12.A	13.A	14.C	15.40	16.4
17.C	18.C	19.C	20.A	21.C	22.D	23.C	24.C
25.C	26.A	27.D	28.4	29.4	30.D	31.D	32.D
33.48	34.8	35.A	36.A	37.A	38.35	39.38	40.A
41.A	42.D	43.A	44.B	45.A	46.A	47.B	48.C
49.B	50.5100	51.3500	52.C	53.A	54.B		

Explanations

1. B

Since, two cars are allowed on each route, The A-M part and N-B will take same time. The difference will be in travelling M-B part and A-N part, and that difference is 0.1 minute. In route M-B car will take $20+0.9 = 20.9$ min and in route A-N car will take $20+1 = 21$ min. So difference = 0.1 min

 VIDEO SOLUTION

2. B

From the previous question we have found that

1 car take AMB route, 2 cars take AMNB route and other take ANB route.

Then the portion A-M will be travelled by 3 cars, M-B by one car, M-N by 2 cars, A-N by 1 car and N-B by 3 cars.

Then travel time of AMB will be $A-M + M-B = (6+3*2) + (20) = 32$

Then travel time of AMNB will be $A-M + M-N + N-B = (6+3*2) + (7+1) + (6+3*2) = 32$

Then travel time of ANB will be $A-N + N-B = (20) + (6+3*2) = 32$

The minimum travel time from A to B is 32 min.

 VIDEO SOLUTION

3. 2

Since there are two routes i.e A-M-B and A-N-B and four cars, then 2 cars must be allowed to take each route. In case if one car tried to break rule, then its travel time will increase. Now assume that on route A-M-B three cars are allowed and on route A-N-B one car is allowed, then one car running on A-M-B can break the rule and reduce its travel time. Hence, two cars must be allowed on each route.

 VIDEO SOLUTION

4. 2

Case 1: Let us assume 1 car takes AMB route, 3 cars take ANB route

Then travel time of AMB will be $A-M + M-B = 6 + 20 = 26$

Then travel time of ANB will be $A-N + N-B = (20+2) + (6+3*2) = 34$

Now, one car(A) travelling on ANB broke the rule and decided to move on AMB route. then

Case 2: Let us assume 2 cars take AMB route, 2 cars take ANB route

Then travel time of AMB will be $A-M + M-B = (6+3) + (20+0.9) = 29.9$

Then travel time of ANB will be $A-N + N-B = (20+1) + (6+3) = 30$

Since the car A reduced its time from 34 to 29.9, the case-1 route is not optimal, hence case 1 is invalid.

Now, one car(B) travelling on ANB broke the rule and decided to move on AMB route. then, case

Case 3: Let us assume 3 cars take AMB route, 1 car take ANB route

Then travel time of AMB will be $A-M + M-B = (6+3*2) + (20+0.9*2) = 33.8$

Then travel time of ANB will be $A-N + N-B = (20) + (6) = 29$

Now, one car(C) travelling on AMB broke the rule and decided to move on AMNB route. then

Case 4: Let us assume 2 cars take AMB route, 1 car take AMNB route and other take ANB route.

Then travel time of AMB will be $A-M + M-B = (6+3*2) + (20+0.9) = 32.9$

Then travel time of AMNB will be $A-M + M-N + N-B = (6+3*2) + (7) + (6+3) = 28$

Then travel time of ANB will be $A-N + N-B = (20) + (6+3) = 29$

Since this car(C) reduced its time from 33.8 to 28, the route is not optimal, hence case 3 is invalid.

Since this car(B) reduced its time from 30 to 28, the route is not optimal, hence case 2 is invalid.

Now, one car(D) travelling on AMB broke the rule and decided to move on AMNB route. then

Case 5: Let us assume 1 car takes AMB route, 2 cars take AMNB route and other takes ANB route.

Then, the portion A-M will be travelled by 3 cars, M-B by one car, M-N by 2 cars, A-N by 1 car and N-B by 3 cars.

Then, travel time of AMB will be $A-M + M-B = (6+3*2) + (20) = 32$

Then, travel time of AMNB will be $A-M + M-N + N-B = (6+3*2) + (7+1) + (6+3*2) = 32$

Then, travel time of ANB will be $A-N + N-B = (20) + (6+3*2) = 32$

It is clear that the car D reduced its time from 32.9 min to 32 min if it broke the rule. Hence, case 4 is invalid.

In this arrangement of case 5, no car can improve their travel time by changing their path

Hence, the optimal allocation will be to order 2 cars on A-M-N-B route.

 VIDEO SOLUTION

5.E

Let the toll charged at junctions A, B, C, and D be a,b,c and d respectively. Now since we want equal traffic through A and C, total cost through routes passing from A and C should be equal. So we have $(9+a+5) + (2+b+2+a+5) = (2+3+b+c+2) + (7+d+1+c+2)$. Only option E satisfy the above equality.

 VIDEO SOLUTION



6.D

Now the fuel cost along different routes are :

SAT = 14

SBAT = 9

SBCT = 7

SDCT = 10

SDT = 13

Now, if we consider option D. Total cost for all routes comes out to be same which is 15. Hence option D.

 VIDEO SOLUTION

7.C

The costs of the routes are as given below:

S - B - C - T = 7

S - B - A - T = 9

S - D - C - T = 10

$$S - D - T = 13$$

$$S - A - T = 14$$

Hence now 100% of the traffic flows through $S - B - C - T$

Now if we make the cost of traveling through $S - B - C - T$ same as some other route not going through B, then the traffic will be equally distributed between these two routes. The lowest such route is $S - D - C - T$. The difference in cost = 3. Hence if we levy a toll of Rs.3 at B, the costs of SBCT and SBAT become 10,12 respectively and other routes are not affected. So 50% traffic flows through SBCT and 50% flows through SDCT. Hence cost in this policy = 10.

 VIDEO SOLUTION

8. **A**

Total cost = fuel cost + toll

Total cost along SAT : $14 + \text{toll}_A$

Total cost along SBAT : $9 + \text{toll}_A + \text{toll}_B$

Total cost along SDT : $13 + \text{toll}_D$

Now when option A is considered, total costs come out to be same.

Hence option A is correct.

 VIDEO SOLUTION

9. **C**

Let the toll charged at junctions A, B, C, and D be a, b, c and d respectively. Then the so that equal amount is collected through all route we have, $9 + a + 5 = 2 + b + 2 + a + 5 = 10 + d + c = 13 + d$. Then from the options only option C satisfies the above equality. hence option C.

 VIDEO SOLUTION

10. **B**

We can start from either city C or D.

C - A - B - D - C - B - E - D

D - B - A - C - B - E - D - C

We cannot start from any other city.

So, we can either start from C or end at C but not both.

 VIEW SOLUTION



CAT WhatsApp Group

cracku



11. **D**

We can start from either city C or D.

C - A - B - D - C - B - E - D

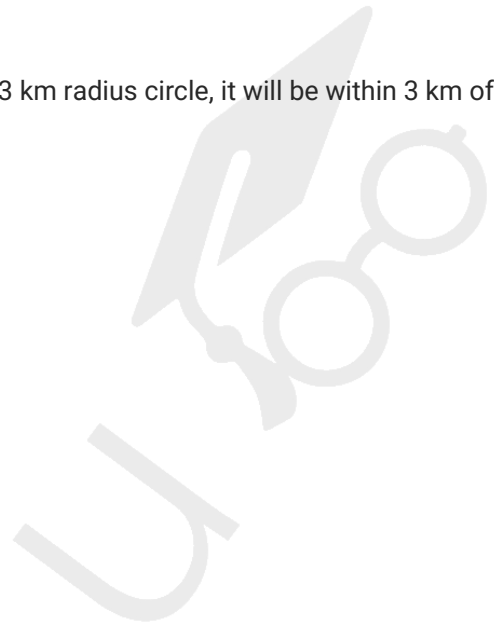
D - B - A - C - B - E - D - C

We cannot start from any other city.

 VIEW SOLUTION

12. **A**

Consider the following scenario:



 [VIEW SOLUTION](#)

Consider the following scenario:

 [VIEW SOLUTION](#)

 VIDEO SOLUTION

36/69

Number of flights between the cities of $G1 = 3 \times 2 \times 4 = 12$

Between cities in A and any city in $G2 = 3 \times 4 = 12$

Between B and any city in $G3 = 2 \times 4 = 8$

Between C and any city in $G4 = 2 \times 4 = 8$

Total = $12 \times 2 + 8 \times 2 = 40$

[VIDEO SOLUTION](#)



16.4

The cities those were a part of $G2$ will be shifted to either $G3$ or $G4$ but that will not have any impact on the number of the total flights from $G1$. The only reduction which will take place due to the number of flights shutting down from A to C.

Hence, the maximum reduction in the number of direct flights as compared to the situation before the operational difficulties arose = 4

Alternate method :

Let us determine the number of flights under new conditions.

Flights between A and B = 4

Between B and any city in $G3 = 4 \times 4 = 16$

Between C and B = 4

Between C and any city in $G4 = 4 \times 4 = 12$

Hence, total flights = 36

Thus, reduction = $40 - 36 = 4$.

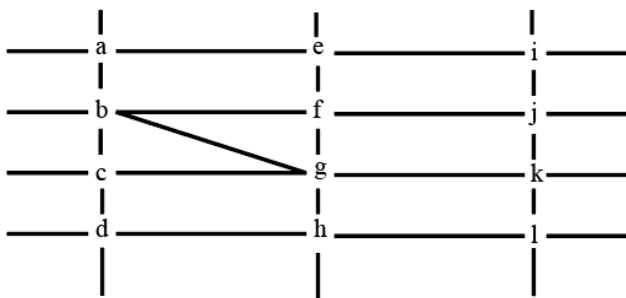
[VIDEO SOLUTION](#)

17.C

There are ten cities. We need to find the minimum number of flights required to travel from any city to any city. Any two cities can be selected in $10C2$ ways. Now for these two cities, a person will need minimum 4 flights. (1 to go from A to B, 1 to go from B to A. Similarly, 1 to return to A and 1 to return to B) Thus, minimum number of required flights = $45 \times 4 = 180$.

[VIDEO SOLUTION](#)

18.C



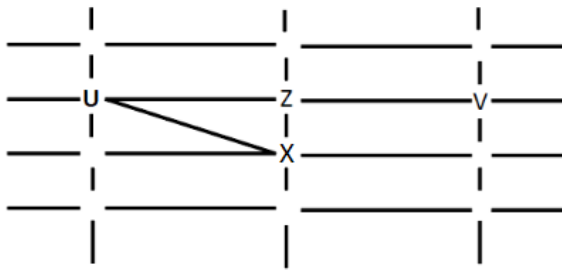
From 1, X, U, and Z are standing at the three corners of a triangle formed by three street segments.

From 2, X can see only U and Z.

From 4, U sees V standing in the next intersection behind Z. Also, no one among the six is standing at intersection d.

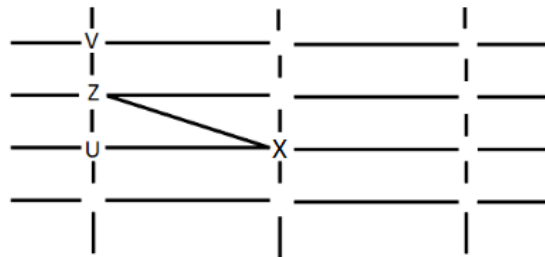
Only cases possible are:

1.



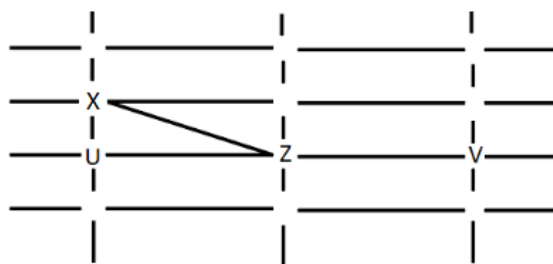
W cannot see V or Z. So W can only be at the intersection a. Since Y can see only U and W, Y can only be at c where X can see him. Hence this case is rejected.

2.



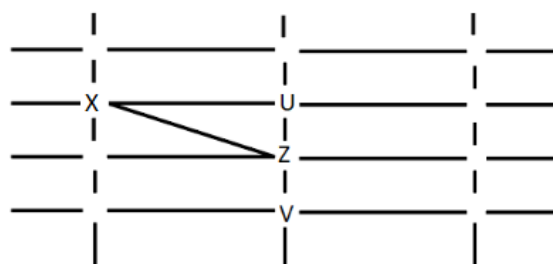
Y can only see U and W. Y cannot be placed anywhere. Hence this case is also rejected.

3.



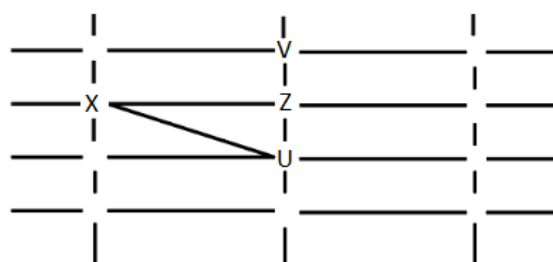
Y can only see U and W. Y cannot be placed anywhere. Hence this case is also rejected.

4.

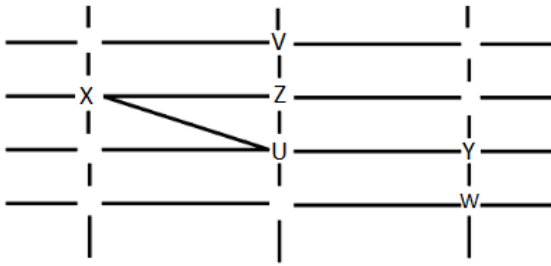


W cannot see V or Z. W can only be placed at i. Y can see only U and W. Y can only be placed at j or e, where he can see more people than U and W. Hence this case is also rejected.

5.



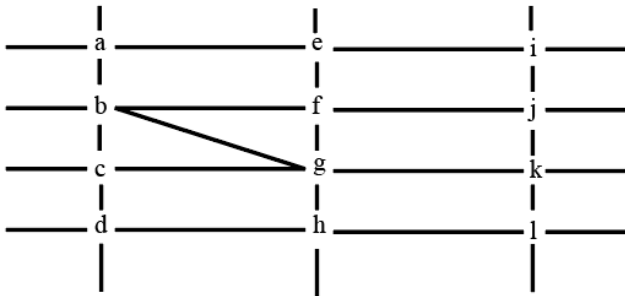
W cannot see V or Z. Y can only see U and W. Hence W and Y can only be placed as shown:



V can see U and Z only. Hence C is the answer.

[VIDEO SOLUTION](#)

19.C



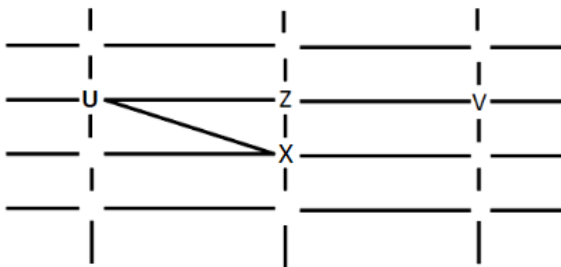
From 1, X, U, and Z are standing at the three corners of a triangle formed by three street segments.

From 2, X can see only U and Z.

From 4, U sees V standing in the next intersection behind Z. Also, no one among the six is standing at intersection d.

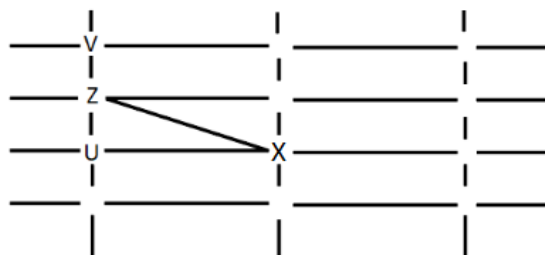
Only cases possible are:

1.



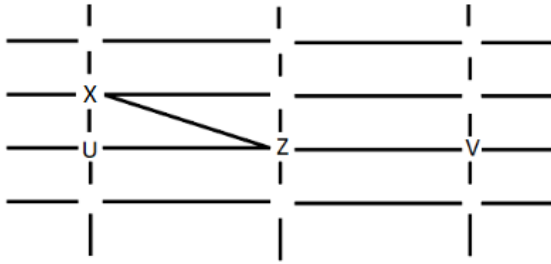
W cannot see V or Z. So W can only be at the intersection a. Since Y can see only U and W, Y can only be at c where X can see him. Hence this case is rejected.

2.



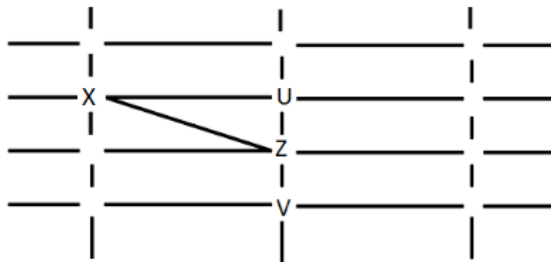
Y can only see U and W. Y cannot be placed anywhere. Hence this case is also rejected.

3.



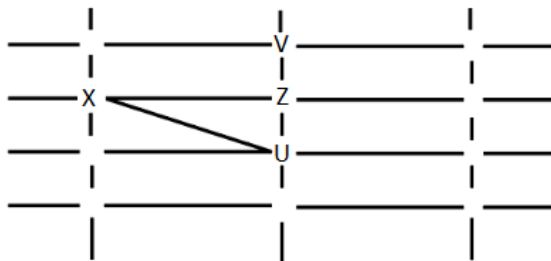
Y can only see U and W. Y cannot be placed anywhere. Hence this case is also rejected.

4.

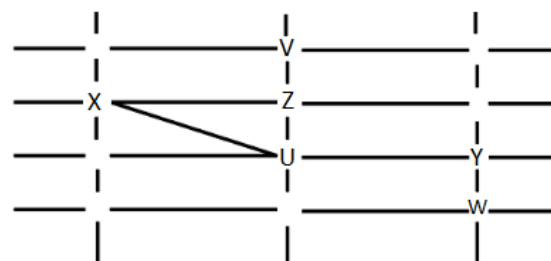


W cannot see V or Z. W can only be placed at i. Y can see only U and W. Y can only be placed at j or e, where he can see more people than U and W. Hence this case is also rejected.

5.



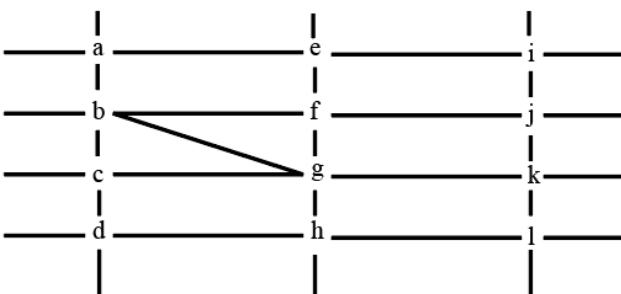
W cannot see V or Z. Y can only see U and W. Hence W and Y can only be placed as shown:



To reach Y, X has to go from b to g and g to k, i.e. 2 streets.

[VIDEO SOLUTION](#)

20. A



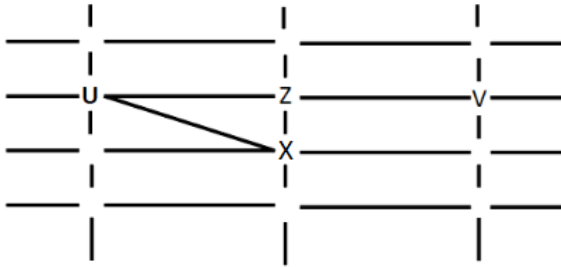
From 1, X, U, and Z are standing at the three corners of a triangle formed by three street segments.

From 2, X can see only U and Z.

From 4, U sees V standing in the next intersection behind Z. Also, no one among the six is standing at intersection d.

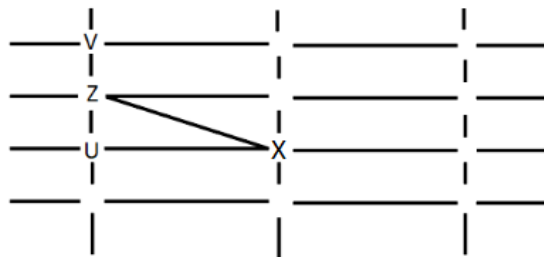
Only cases possible are:

1.



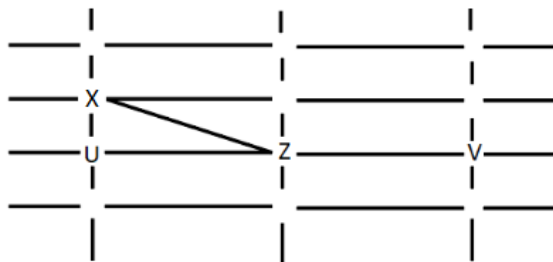
W cannot see V or Z. So W can only be at the intersection a. Since Y can see only U and W, Y can only be at c where X can see him. Hence this case is rejected.

2.



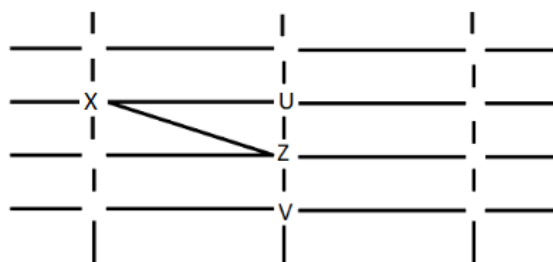
Y can only see U and W. Y cannot be placed anywhere. Hence this case is also rejected.

3.



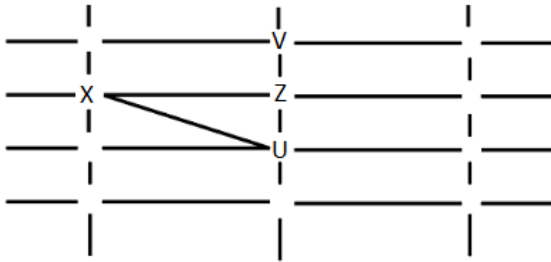
Y can only see U and W. Y cannot be placed anywhere. Hence this case is also rejected.

4.

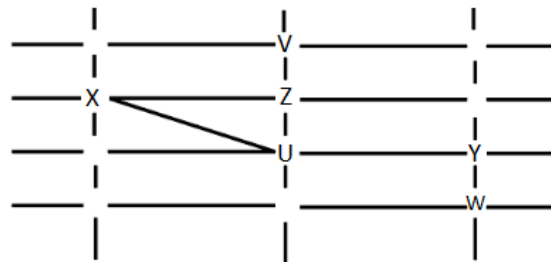


W cannot see V or Z. W can only be placed at i. Y can see only U and W. Y can only be placed at j or e, where he can see more people than U and W. Hence this case is also rejected.

5.



W cannot see V or Z. Y can only see U and W. Hence W and Y can only be placed as shown:



If a new person stands at d(left down corner), they can see W and X only. Hence A is the answer.

VIDEO SOLUTION

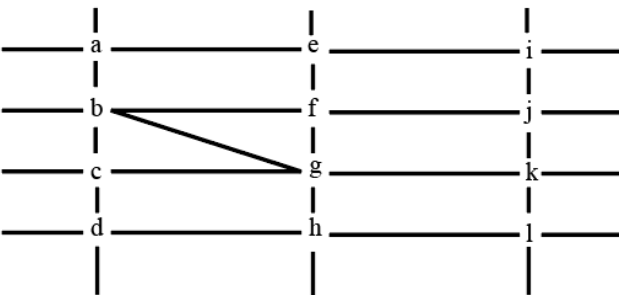


CAT Online Coaching

your road to IIM stars here



21. C



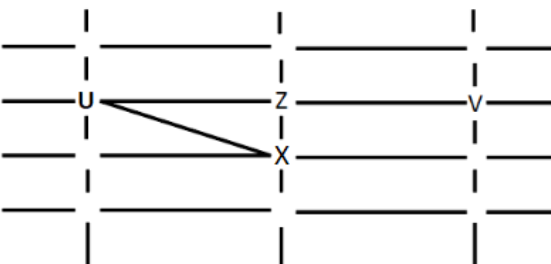
From 1, X, U, and Z are standing at the three corners of a triangle formed by three street segments.

From 2, X can see only U and Z.

From 4, U sees V standing in the next intersection behind Z. Also, no one among the six is standing at intersection d.

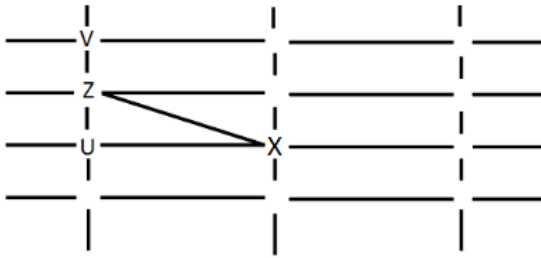
Only cases possible are:

1.



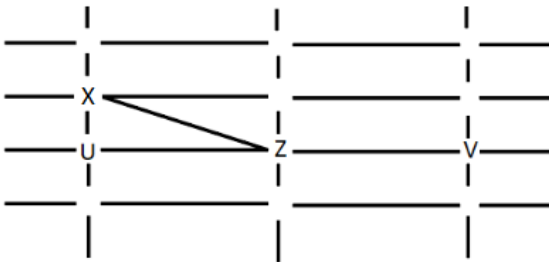
W cannot see V or Z. So W can only be at the intersection a. Since Y can see only U and W, Y can only be at c where X can see him. Hence this case is rejected.

2.



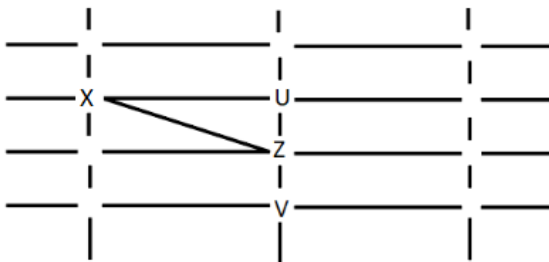
Y can only see U and W. Y cannot be placed anywhere. Hence this case is also rejected.

3.



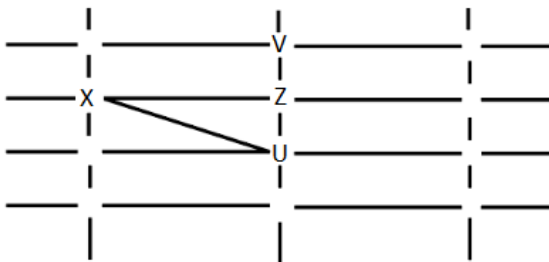
Y can only see U and W. Y cannot be placed anywhere. Hence this case is also rejected.

4.

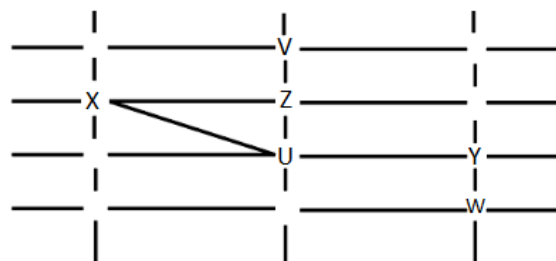


W cannot see V or Z. W can only be placed at i. Y can see only U and W. Y can only be placed at j or e, where he can see more people than U and W. Hence this case is also rejected.

5.



W cannot see V or Z. Y can only see U and W. Hence W and Y can only be placed as shown:



No one is standing at the intersection A. Hence C is the answer.

[VIDEO SOLUTION](#)

22. D

Option (a): When queen is at f8. In this case h8 and b4 will be under attack.

	a	b	c	d	e	f	g	h
8						(f8)		h8
7				d7				h7
6								
5								
4		b4						
3	a3							
2								
1	a1							

Option (b): When queen is at a7. In this case a3 and d7 will be under attack.

	a	b	c	d	e	f	g	h
8								h8
7	(a7)			d7				h7
6								
5								
4		b4						
3	a3							
2								
1	a1							

Option (c): When queen is at c1. In this case a1 and a3 will be under attack.

	a	b	c	d	e	f	g	h
8								h8
7				d7				h7
6								
5								
4		b4						
3	a3							
2								
1	a1		(c1)					

Option (d): When queen is at d3. In this case a3, d7 and h3 will be under attack.

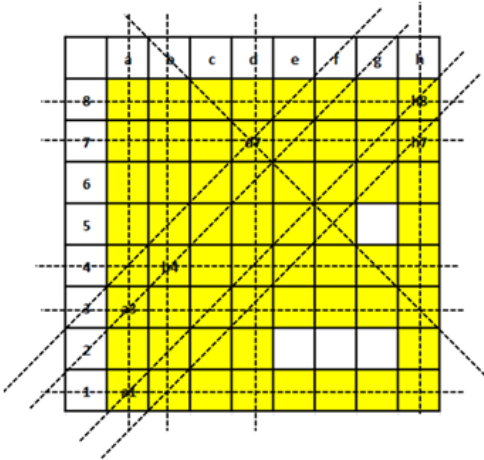
	a	b	c	d	e	f	g	h
8								h8
7				d7				h7
6								
5								
4		b4						
3	a3			(d3)				
2								
1	a1							

Therefore, we can say that option D is the correct answer.

VIDEO SOLUTION

23. C

From the diagram we can see that except positions e2, f2, g2 and g5 queen can attack at least one among the given pieces.



Hence, we can say that there are exactly for position from where queen can't attack any of the given pieces. Therefore, option C is the correct answer.

[VIDEO SOLUTION](#)

24. C

Let us draw the diagram and mark position of various pieces as given in the question.

	a	b	c	d	e	f	g	h
8								
7								
6								
5			Queen				g5	
4								
3	a3						g3	
2			c2					
1							g1	

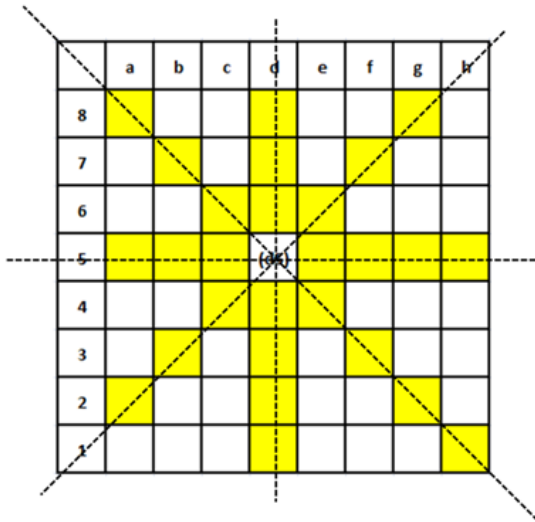
Attack line is shown by the yellow color. All the pieces on this line will be under attack.

	a	b	c	d	e	f	g	h
8								
7								
6								
5			Queen				g5	
4								
3	a3						g3	
2			c2					
1							g1	

From the diagram we can see that a3, g1, c2 and g5 are under attack. Hence, option C is the correct answer.

[VIDEO SOLUTION](#)

25. C



From the diagram we can see that the number of positions those are safe from queen's attack = $6 + 5 + 5 + 5 + 5 + 5 + 5 = 36$. Therefore, option C is the correct answer.

[VIDEO SOLUTION](#)



26. A

Let us consider a 5x5 matrix. Let us start with the top left square and fill number 1 in as many squares as possible.

1		1		1
1		1		1
1		1		1

We have to use a second number, 2 to fill the gap between two 1s.

1	2	1	2	1
1	2	1	2	1
1	2	1	2	1

All the cells in row 2 and row 4 are adjacent to the cells containing numbers 1 and 2. Therefore, rows 2 and 4 should be filled with a new set of numbers. We need at least 2 numbers to fill a row such that the adjacent cells do not contain the same number (by alternating the numbers in the consecutive cells). Rows 2 and 4 are completely isolated from each other and hence, the same set of numbers can be used to fill both the rows.

1	2	1	2	1
3	4	3	4	3
1	2	1	2	1
3	4	3	4	3
1	2	1	2	1

4 numbers are required to fill a 5x5 matrix.

It has been given that we are allowed to make 1 mistake - One pair of adjacent cells can contain the same number. In the arrangement given above, we can alter any value along the edge to satisfy this condition. For example, the 2 in the bottom-most row can be changed to 4. Still, the number of numbers required to fill the matrix will be 4.

Another way to approach this problem is as follows:

We know that a minimum of 4 numbers are required to fill a 5x5 matrix. If we are allowed to make a mistake, then the number of numbers required should either remain the same or go down. 4 is the smallest value among the given options. Therefore, we can be sure that even if we are allowed to make a mistake, 4 numbers will be required to fill the matrix and hence, option A is the right answer.

 VIDEO SOLUTION

27.D

It has been given that all the cells adjacent to a cell must have different numerals. Let us start filling the matrix from the central square since the central square has the maximum number of squares adjacent to it (8) and it will be easier to work around the central 9 squares. A minimum of 9 numbers will be required to fill the central 9 squares.

	2	3	4	
	9	1	5	
	8	7	6	

Now we have to fill the remaining squares. Let us start with the top left square. We have to check whether the 9 numbers will be sufficient to fill all the squares such that no 2 squares adjacent to a square have the same number. We can use any of the 3 numbers 4, 5, and 6 to fill the top left square since none of the numbers in the second column are adjacent to these numbers.

Let us assume that we use 4 to fill the top left square. Now, one of the cells with the number 4 has become adjacent to the cell with number 2 and no other cell adjacent to cell with number 2 (in the second row and second column) can have 4 as its neighbour. Similarly, we can fill the first row with numbers 8 and 7.

4	7	8		
5	2	3	4	
6	9	1	5	
	8	7	6	

In essence, we are trying to create a grid around each of the numbers in the corners of the inner 3x3 matrix such that no 2 cells adjacent to a cell have the same number. Filling the other cells similarly, we get the following matrix as one of the possible cases.

4	7	8	6	7
5	2	3	4	9
6	9	1	5	2
3	8	7	6	8
2	5	4	3	9

We need a minimum of 9 numbers to fill a 5x5 matrix such that for any cell, no 2 cells adjacent to it contain the same value. Therefore, option D is the right answer.

 VIDEO SOLUTION

28.4

Let us consider a 5x5 matrix. Let us start with the top left square and fill number 1 in as many squares as possible.

1		1		1
1		1		1
1		1		1

We have to use a second number, 2 to fill the gap between two 1s.

1	2	1	2	1
1	2	1	2	1
1	2	1	2	1

All the cells in row 2 and row 4 are adjacent to the cells containing numbers 1 and 2. Therefore, rows 2 and 4 should be filled with a new set of numbers. We need at least 2 numbers to fill a row such that the adjacent cells do not contain the same number (by alternating the numbers in the consecutive cells). Rows 2 and 4 are completely isolated from each other and hence, the same set of numbers can be used to fill both the rows.

1	2	1	2	1
3	4	3	4	3
1	2	1	2	1
3	4	3	4	3
1	2	1	2	1

As we can see, a minimum of 4 numbers are required to fill a 5x5 matrix. Therefore, 4 is the correct answer.

 VIDEO SOLUTION

29.4

Let us use 1 to denote the first number that we fill. We have to fill as many squares with 1 as possible. If we start with the top-left square, we can fill 4 squares with the number 1.

1		1
1		1

Now, we can fill number 2 only in 2 of the 5 squares available.

1	2	1
1	2	1

The 3 squares available now are adjacent to each other. Therefore, we will require at least 2 numbers to fill these squares.

1	2	1
3	4	3
1	2	1

We need a minimum of 4 numbers to fill a 3x3 square matrix such that no 2 adjacent cells contain the same number. Therefore, 4 is the correct answer.

[VIDEO SOLUTION](#)

30. D

We know that quantity between Vaishali and jyotishmati is 300,

So quantity in avanti-vaishal route should be $400+300=700$.

So free capacity is $1000-700=300$. Hence option D .

[VIDEO SOLUTION](#)



CAT Syllabus PDF



31. D

We know that quantity between Vaishali and jyotishmati is 300, So quantity in avanti-vaishal route should be 700. Now at jyotishmati the required quantity is $400+700 = 1100$.

But through vaishali only 300 comes , so 800 should come through Vidisha-jyotishmati route.

Now demand at vidisha is 200. So total quantity required in avanti - vidisha route is $800+200=1000$.

Hence, free capacity available = $1000-1000=0$. Hence option D.

[VIDEO SOLUTION](#)

32. D

We know that quantity between Vaishali and jyotishmati is 300,

So quantity in avanti-vaishal route should be 700.

Now at jyotishmati the required quantity is $400+700 = 1100$.

But through vaishali only 300 comes , so 800 should come through Vidisha-jyotishmati route.

Now demand at vidisha is 200. So total quantity required in avanti - vidisha route is $800+200=1000$. Hence option D.

[VIDEO SOLUTION](#)

33. 48

Travel time between A and B = $(10 \times 3) + (7 \times 1) + (2 \times 2) = 41$ minutes

After completing a journey, a train must rest for 15 minutes at least before starting again.

So if a train starts from 6 am from A to B, then the latest by which that train will start from B to A will be at 7 am, as in the north-south direction, a train starts from A and B every 15 minutes.

So the total no. of trains required for the north-south lines = $\left(\frac{60}{15}\right) \times 2 \times 2 = 16$

Travel time between M and N = $(20 \times 2) + (17 \times 1) + (2 \times 2) = 61$ minutes

After completing a journey, a train must rest for 15 minutes at least before starting again.

So if a train starts from 6 am from M to N, then the latest by which that train will start from N to M will be at 7:20 am, as in the east-west direction, a train starts from M and N every 15 minutes.

So the total no. of trains required for the east-west lines = $\left(\frac{80}{10}\right) \times 2 \times 2 = 32$

Total no. of trains required to service the city = $16 + 32 = 48$

 VIDEO SOLUTION

34.8

Travel time between A and B = $(10 \times 3) + (7 \times 1) + (2 \times 2) = 41$ minutes

After completing a journey, a train must rest for 15 minutes at least before starting again.

So if a train starts from 6 am from A to B, then the latest by which that train will start from B to A will be at 7 am, as in the north-south direction, a train starts from A and B every 15 minutes.

So the total no. of trains required = $\left(\frac{60}{15}\right) \times 2 = 8$

 VIDEO SOLUTION

35. A

In the east-west direction, a train starts from station M every 10 minutes.

Now the first train leaving station M is at 6:00 am.

Since every 10 minutes trains are available in east-west direction, the train timings leaving from M will be like 6:10 am, 6:20 am, 6:30 am and so on.

Given, Hari has reached station M by 8:05 am.

So the earliest by which Hari can catch a train from station M is 8:10 am.

Now there are 19 stations between M and n, out of which two stations are junctions.

Time taken to travel between two stations in the east-west direction is 2 minutes.

Therefore, the time for which the train was running between M and N (excluding the stoppage time) = $20 \times 2 = 40$ minutes

Stoppage time at a junction is 2 minutes, while at the rest of the stations, it is 1 minute each.

Stoppage time for the train running between M and N = $(17 \times 1) + (2 \times 2) = 21$ minutes

Therefore, total travel time = $40 + 21 = 61$ minutes.

So the time by which Hari reaches N is 8:10 am + 61 minutes = 9:11 am

 VIDEO SOLUTION



36. A

Priya can reach S from T via R or V.

Case 1:- T-V-S

In the east-west direction, the first train from P arrives at T at time = 6 am + $(4 \times 2) + (3 \times 1) = 11$ minutes = 6:11 am

Since T is a junction so this train will halt for 2 minutes at T and leave at 6:13.

Priya boards a east-west train then Priya will board a train for V from T at 10:33 am.

There are 9 stations between T and V

Travelling time between T and V = $(10 \times 2) + (9 \times 1) = 29$ minutes

Therefore, Priya will reach V latest by 10:33 am + 29 minutes = 11:02 am

In the north-south direction, the first train from D arrives at V at time = 6 am + $(3 \times 3) + (2 \times 1) = 11$ minutes = 6:11 am

Since V is a junction so this train will halt for 2 minutes at V and leave at 6:13.

Since every 15 minutes, a train starts from D in the north-south direction, so the latest by which Priya will be able to board such a train from V is at 11:13 am.

There are 3 stations between V and S

Travelling time between R and S = $(4 \times 3) + ((3 \times 1)) = 15$ minutes

Time by which she reaches S = 11:13 + 15 minutes = 11:28 am

Case 2:- T-R-S

In the north-south direction, the first train from B arrives at T at time = 6 am + $(3 \times 3) + (2 \times 1) = 11$ minutes = 6:11 am

Since T is a junction so this train will halt for 2 minutes at T and leave at 6:13.

Since every 15 minutes a train starts from P in the east-west direction so the latest by which Priya will be able to board such a train is at 10:28 am.

Now since she will be able to board a north-south train earlier than the east-west train so Priya will board a train for R from T at 10:28 am.

There are 3 stations between T and R

Travelling time between T and R = $(4 \times 3) + ((3 \times 1)) = 15$ minutes

Therefore, Priya will reach R latest by 10:43 am

In the east-west direction, the first train from M arrives at R at time = 6 am + $(4 \times 2) + (3 \times 1) = 11$ minutes = 6:11 am

Since R is a junction so this train will halt for 2 minutes at R and leave at 6:13.

Since every 10 minutes, a train starts from M in the east-west direction, so the latest by which Priya will be able to board such a train is at 10:43 am.

There are 9 stations between R and S

Travelling time between R and S = $(10 \times 2) + (9 \times 1) = 29$ minutes

Time by which she reaches S = 10:43 + 29 minutes = 11:12 am

We are getting shorter time in case 2. So 11:12 am is the answer

 VIDEO SOLUTION

37. A

Travelling time between S and R = $(10 \times 2) + (9 \times 1) = 29$ minutes

There is a stoppage of 2 minutes at R

Travelling time between R and B = $(7 \times 3) + (1 \times 2) + (5 \times 1) = 28$ minutes

In the north-south direction, the first train from A arrives at R at time = 6 am + $(3 \times 3) + (2 \times 1) = 6:11$ am.

Since R is a junction so this train will halt for 2 minutes at R and leave at 6:13.

Every 15 minutes, a train starts from A in the north-south direction.

The last train that leaves A will be at 12:00 am and it will leave R at 12:13 am, so Haripriya must reach R till 12:13 am.

Travelling time between S and R = $(10 \times 2) + (9 \times 1) = 29$ minutes

So Haripriya must board the train at S by 11:44 pm

In the east-west direction, the first train from N arrives at S at time = 6 am + $(6 \times 2) + (5 \times 1) = 6:17$ am.

Since S is a junction so this train will halt for 2 minutes at S and leave at 6:19.

Every 10 minutes, a train starts from N in the east-west direction.

Therefore, Haripriya should board the train which leaves S at 11:39.

 VIDEO SOLUTION

38.35

Points to be noted:

1. Starts from the warehouse and ends at a location after visiting all the locations exactly once.
2. While making the route plan, the supplier goes to the locations in decreasing order of demand. If equal demand, goes to the nearest ones first.
3. While creating the route, the supplier can either follow the direct path (if available) from one location to another or can take the path via the warehouse (Prefers minimum distance).

In the question, it is given that last location is A. The demand in the remaining places should be greater than A. This implies A demand cannot be 70 units. Therefore, it is 50 units.

The demand of the location D is 30 or 50 units. This implies this should be placed before D.

The demand of the location placed before D should be greater than or equal to 50 units. Location supplier visited before D is B (60 units of demand). It cannot be C because values of C is greater than the values of B.

Therefore, order is C - B - D - A.

From warehouse to C - 12 km

C to B - 4 km

B to D - 12 km

D to A - 7 km (through warehouse)

Total distance covered = $12 + 4 + 12 + 7 = 35$ km

This question is removed from the paper because if the order is CBDA, the supplier will go to A from B (and hence the last city visited will be D).

 VIDEO SOLUTION

39.38

Points to be noted:

1. Starts from the warehouse and ends at a location after visiting all the locations exactly once.
2. While making the route plan, the supplier goes to the locations in decreasing order of demand. If equal demand, goes to the nearest ones first.
3. While creating the route, the supplier can either follow the direct path (if available) from one location to another or can take the path via the warehouse (supplier prefers minimum distance).

In the question, it is given that total number of units delivered is 250 units.

Maximum number of widgets that can be delivered is 70 units(A) + 50 units(D) + 60 units(B) + 100 units(C) = 280 units

From this 30 units should be decreased. 30 units can be decreased only when C's demand decreases to 70 units (because the difference for remaining locations is 20 units)

Therefore, the only possibility is 70 units(A) + 50 units(D) + 60 units(B) + 70 units(C)

From statement 1, the order should be A - C - B - D (as A is nearer to warehouse than C)

Distance from Warehouse to A is 5 km

Distance from A to C is 17 km

Distance from C to B is 4 km

Distance from B to D is 12 km

Total distance covered = $5 + 17 + 4 + 12 = 38$ km

 VIDEO SOLUTION

40. **A**

It is given that the first location visited from the warehouse is A.

If A's demand is 50 units, C's demand should be less than 50 units which is not possible. This implies demand of locations A and C is 70 units.

A(70 units) $\rightarrow$ C(70 units)

Warehouse to A - 5km

A to C - 17 km

Distance covered = $5 + 17 = 22$ km

Remaining distance = $40 - 22 = 18$ km

C to B - 4 km

B to D - 12 km

Distance covered = $4 + 12 = 16$ km $\neq 18$ km

C to D - 6 km

D to B - 12 km

Distance covered = $6 + 12 = 18$ km

Therefore, supplier can cover distance 18 km if he visits D before B, i.e. demand of D should be more than demand of B. This is only possible when D's demand is 50 units and B's demand is 40 units.

It is given,

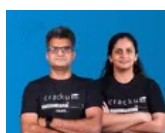
D - 50 units - 60% probability

B - 40 units - 30% probability

Required value = $0.6 \times 0.3 = 0.18 = 18\%$

The answer is option A.

 VIDEO SOLUTION

**IIM Call Predictor**
Accurate analysis of your profile



41. **A**

Points to be noted:

1. Starts from the warehouse and ends at a location after visiting all the locations exactly once.
2. While making the route plan, the supplier goes to the locations in decreasing order of demand. If equal demand, goes to the nearest ones first.
3. While creating the route, the supplier can either follow the direct path (if available) from one location to another or can take the path via the warehouse(Prefers minimum distance).

Demand in all other locations should be less than or equal to the demand in first location. In the question, it is given that A is not first location. B and D cannot be first location. This implies C should be the first location.

It is given, total route distance is 29 km.

Warehouse to C is 12 km.

This implies remaining distance should be $29 - 12$, i.e. 17km

This is only possible when C visits B, A and D later(4 + 6 + 7).

The order should be C - B - A - D

C's demand can be 70/100, B's demand should be 60, A's demand should be 50 and D's demand can be 30/50.

The possible number of widgets delivered can be

$$70 + 60 + 50 + 30 = 210$$

$$70 + 60 + 50 + 50 = 230$$

$$100 + 60 + 50 + 30 = 240$$

$$100 + 60 + 50 + 50 = 260$$

The answer is option A.

 VIDEO SOLUTION

42. **D**

Maximum number of widgets delivered in a day is 100 units(C) + 70 units(A) + 60 units(B) + 50 units(D), i.e. 280 units

Given, total number of widgets delivered is 260 units. This implies 20 units must be decreased from any one of the locations.

A - (70,50), B - (60,40), C - (100,70) and D - (50,30)

20 units can be decreased from A, B or D.

Demand at location C will be 100 units and supplier first visits C.

In the question, it is also given that the route ends at B.

C(100 units), →, →, B

If B's demand is 60 units, D's demand should be more than 60 units which is not possible. Therefore, B's demand should be 40 units.

20 units is decreased at location B. This implies demand at location A is 70 units and at location D is 50 units.

Order will be C(100 units) - A(70 units) - D(50 units) - B(40 units).

It is given,

C - 100 units - 70%

A - 70 units - 60%

D - 50 units - 60%

B - 40 units - 30%

$$\text{Required value} = 0.7 \times 0.6 \times 0.6 \times 0.3 = 0.0756 = 7.56\%$$

The answer is option D.

 VIDEO SOLUTION

43. **A**

There are 12 plots and each of them got even number of plots. So, possible cases are 4,4,2,2 or 6,2,2,2.

From 4, A and B got more plots than D. So, the only possible case is A, B each got 4 and C, D each got 2.

From 6, D has to get two adjacent plots and From 8, plots of C, D are not adjacent to each other => D must have got plots in X3, X4.

C already has two plots in X1, Z2. So, the corner plot Z4 should belong to B.

From 7, B has a plot in each row and each column. So, X2 should belong to B.

Now, out of the remaining Y2, Y3, Y4 and Z3 three plots belong to A and one belongs to B.

Till now B hasn't got any plot in Third column and 2nd row.

So, Y3 belongs to B and Y2, Y4, Z3 belongs to A.

Let the number of trees in Y4 be $4x$ from 3, number of trees in Y3, Y2 will be $2x, x$ respectively.

The number of teak trees = $7x + 21$

∴ Number of mango trees = $14x + 42$

The table now looks like:

	1	2	3	4	
X	12 C				$14x + 42$
Y	21 A	x A	$2x$ B	$4x$ A	$7x + 21$
Z			9 A	28 B	

Each plot had trees in non-zero multiples of 3 or 4 and none of the plots had the same number of trees and from 2, B didn't have the largest number of trees in a plot $\Rightarrow x < 8$.

x can't be 7, 5, 3, 2, 1 as for these cases at least one of $x, 2x, 4x$ is neither multiple of 3 or 4.

x can be 6 or 4.

If $x = 6$, number of Teak trees will be 63 and Mango trees will be 126 $\Rightarrow$ Number of Pine trees = $205 - 126 - 63 = 16$ but number of trees in $Z3 + Z4 > 16$ so, $x \neq 6$.

If $x = 4$, Number of Teak trees = 49 and Mango trees = 98 $\Rightarrow$ Number of Pine trees = 58. Valid case.

Number of trees with A = $30 + 5x = 50$.

From 1, number of trees with C, D = 30, 56 respectively.

So, number of trees in $Z2 = 18$.

∴ Number of trees with B = $205 - 50 - 30 - 56 = 69$.

From 2, largest number of trees in a plot is 32. They can be in the plot of either B or D. If they are from B, they have to be from $X2$ but in that case number of trees in $Z1 = 1$ which is neither a multiple of 3 or 4.

So, highest number of trees in a plot are with D and it is 32 $\Rightarrow$ number of trees in $X3, X4$ are 32, 24 in any order.

So, number of trees in $X2 = 98 - 56 - 12 = 30$

∴ Number of trees in $Z1 = 69 - 30 - 28 - 8 = 3$.

The final table will look like:

	1	2	3	4	
X	12 C	30 B	32/24 D	24/32 D	98
	21 A	4 A	8 B	16 A	49
Y	3 B	18 C	9 A	28 B	58
Z					

Column 1,2,3,4 have 36, 52, 49, 68 trees respectively.

Hence A is correct answer.

[VIDEO SOLUTION](#)

44. B

There are 12 plots and each of them got even number of plots. So, possible cases are 4,4,2,2 or 6,2,2,2.

From 4, A and B got more plots than D. So, the only possible case is A, B each got 4 and C,D each got 2.

From 6, D has to get two adjacent plots and From 8, plots of C, D are not adjacent to each other $\Rightarrow$ D must have got plots in X3, X4.

C already has two plots in X1, Z2. So, the corner plot Z4 should belong to B.

From 7, B has a plot in each row and each column. So, X2 should belong to B.

Now, out of the remaining Y2, Y3, Y4 and Z3 three plots belong to A and one belongs to B.

Till now B hasn't got any plot in Third column and 2nd row.

So, Y3 belongs to B and Y2, Y4, Z3 belongs to A.

Let the number of trees in Y4 be $4x$ from 3, number of trees in Y3, Y2 will be $2x, x$ respectively.

The number of teak trees $= 7x + 21$

$\therefore$ Number of mango trees $= 14x + 42$

The table now looks like:

	1	2	3	4	
X	12 C				$14x + 42$
	21 A	x A	2x B	4x A	$7x + 21$
Y			9 A	28 B	
Z					

Each plot had trees in non-zero multiples of 3 or 4 and none of the plots had the same number of trees and from 2, B didn't have the largest number of trees in a plot $\Rightarrow x < 8$.

x can't be 7, 5, 3, 2, 1 as for these cases at least one of $x, 2x, 4x$ is neither multiple of 3 or 4.

x can be 6 or 4.

If $x=6$, number of Teak trees will be 63 and Mango trees will be 126 $\Rightarrow$ Number of Pine trees = $205 - 126 - 63 = 16$ but number of trees in $Z_3 + Z_4 > 16$ so, $x \neq 6$.

If $x=4$, Number of Teak trees = 49 and Mango trees = 98 $\Rightarrow$ Number of Pine trees = 58. Valid case.

Number of trees with A = $30 + 5x = 50$.

From 1, number of trees with C, D = 30, 56 respectively.

So, number of trees in $Z_2 = 18$.

$\therefore$ Number of trees with B = $205 - 50 - 30 - 56 = 69$.

From 2, largest number of trees in a plot is 32. They can be in the plot of either B or D. If they are from B, they have to be from X_2 but in that case number of trees in $Z_1 = 1$ which is neither a multiple of 3 or 4.

So, highest number of trees in a plot are with D and it is 32 $\Rightarrow$ number of trees in X_3, X_4 are 32, 24 in any order.

So, number of trees in $X_2 = 98 - 56 - 12 = 30$

$\therefore$ Number of trees in $Z_1 = 69 - 30 - 28 - 8 = 3$.

The final table will look like:

	1	2	3	4	
X	12 C	30 B	32/24 D	24/32 D	98
Y	21 A	4 A	8 B	16 A	49
Z	3 B	18 C	9 A	28 B	58

$\therefore$ Number of trees per plot is least for Benna = 3.

 VIDEO SOLUTION

45. A

There are 12 plots and each of them got even number of plots. So, possible cases are 4, 4, 2, 2 or 6, 2, 2, 2.

From 4, A and B got more plots than D. So, the only possible case is A, B each got 4 and C, D each got 2.

From 6, D has to get two adjacent plots and From 8, plots of C, D are not adjacent to each other $\Rightarrow$ D must have got plots in X_3, X_4 .

C already has two plots in X_1, Z_2 . So, the corner plot Z_4 should belong to B.

From 7, B has a plot in each row and each column. So, X_2 should belong to B.

Now, out of the remaining Y_2, Y_3, Y_4 and Z_3 three plots belong to A and one belongs to B.

Till now B hasn't got any plot in Third column and 2nd row.

So, Y_3 belongs to B and Y_2, Y_4, Z_3 belongs to A.

Let the number of trees in Y_4 be $4x$ from 3, number of trees in Y_3, Y_2 will be $2x, x$ respectively.

The number of teak trees= $7x+21$

∴ Number of mango trees= $14x+42$

The table now looks like:

	1	2	3	4	
X	12 C				$14x+42$
Y	21 A	x A	2x B	4x A	$7x+21$
Z			9 A	28 B	

Each plot had trees in non-zero multiples of 3 or 4 and none of the plots had the same number of trees and from 2, B didn't have the largest number of trees in a plot $\Rightarrow x < 8$.

x can't be 7,5,3,2,1 as for these cases at least one of x,2x,4x is neither multiple of 3 or 4.

x can be 6 or 4.

If $x=6$, number of Teak trees will be 63 and Mango trees will be 126 $\Rightarrow$ Number of Pine trees= $205-126-63=16$ but number of trees in $Z_3+Z_4 > 16$ so, $x \neq 6$.

If $x=4$, Number of Teak trees=49 and Mango trees=98 $\Rightarrow$ Number of Pine trees=58. Valid case.

Number of trees with A= $30+5x=50$.

From 1, number of trees with C, D= 30, 56 respectively.

So, number of trees in $Z_2= 18$.

∴ Number of trees with B= $205-50-30-56=69$.

From 2, largest number of trees in a plot is 32. They can be in the plot of either B or D. If they are from B, they have to be from X_2 but in that case number of trees in $Z_1=1$ which is neither a multiple of 3 or 4.

So, highest number of trees in a plot are with D and it is 32 $\Rightarrow$ number of trees in X_3, X_4 are 32, 24 in any order.

So, number of trees in $X_2= 98-56-12=30$

∴ Number of trees in $Z_1=69-30-28-8=3$.

The final table will look like:

	1	2	3	4	
X	12 C	30 B	32/24 D	24/32 D	98
Y	21 A	4 A	8 B	16 A	49
Z	3 B	18 C	9 A	28 B	58

Number of Pine trees received by Chitra = 18.

[VIDEO SOLUTION](#)

CAT Percentile Predictor



46. A

There are 12 plots and each of them got even number of plots. So, possible cases are 4,4,2,2 or 6,2,2,2.

From 4, A and B got more plots than D. So, the only possible case is A, B each got 4 and C,D each got 2.

From 6, D has to get two adjacent plots and From 8, plots of C, D are not adjacent to each other $\Rightarrow$ D must have got plots in X3, X4.

C already has two plots in X1, Z2. So, the corner plot Z4 should belong to B.

From 7, B has a plot in each row and each column. So, X2 should belong to B.

Now, out of the remaining Y2, Y3, Y4 and Z3 three plots belong to A and one belongs to B.

Till now B hasn't got any plot in Third column and 2nd row.

So, Y3 belongs to B and Y2, Y4, Z3 belongs to A.

Let the number of trees in Y4 be $4x$ from 3, number of trees in Y3, Y2 will be $2x, x$ respectively.

The number of teak trees = $7x + 21$

$\therefore$ Number of mango trees = $14x + 42$

The table now looks like:

	1	2	3	4	
X	12 C				$14x+42$
Y	21 A	x A	2x B	4x A	$7x+21$
Z			9 A	28 B	

Each plot had trees in non-zero multiples of 3 or 4 and none of the plots had the same number of trees and from 2, B didn't have the largest number of trees in a plot $\Rightarrow x < 8$.

x can't be 7,5,3,2,1 as for these cases at least one of x,2x,4x is neither multiple of 3 or 4.

x can be 6 or 4.

If $x=6$, number of Teak trees will be 63 and Mango trees will be 126 $\Rightarrow$ Number of Pine trees = $205-126-63=16$ but number of trees in $Z_3+Z_4 > 16$ so, $x \neq 6$.

If $x=4$, Number of Teak trees=49 and Mango trees=98 $\Rightarrow$ Number of Pine trees=58. Valid case.

Number of trees with A = $30+5x=50$.

From 1, number of trees with C, D = 30, 56 respectively.

So, number of trees in $Z_2 = 18$.

$\therefore$ Number of trees with B = $205-50-30-56=69$.

From 2, largest number of trees in a plot is 32. They can be in the plot of either B or D. If they are from B, they have to be from X_2 but in that case number of trees in $Z_1=1$ which is neither a multiple of 3 or 4.

So, highest number of trees in a plot are with D and it is 32 $\Rightarrow$ number of trees in X_3, X_4 are 32, 24 in any order.

So, number of trees in $X_2 = 98-56-12=30$

$\therefore$ Number of trees in $Z_1 = 69-30-28=9$.

The final table will look like:

	1	2	3	4	
X	12 C	30 B	32/24 D	24/32 D	98
Y	21 A	4 A	8 B	16 A	49
Z	3 B	18 C	9 A	28 B	58

Sequence of trees received by Abha, Bina, Chitra and Dipti is 50,69,30,56.

47. B

There are 12 plots and each of them got even number of plots. So, possible cases are 4,4,2,2 or 6,2,2,2.

From 4, A and B got more plots than D. So, the only possible case is A, B each got 4 and C,D each got 2.

From 6, D has to get two adjacent plots and From 8, plots of C, D are not adjacent to each other => D must have got plots in X3, X4.

C already has two plots in X1, Z2. So, the corner plot Z4 should belong to B.

From 7, B has a plot in each row and each column. So, X2 should belong to B.

Now, out of the remaining Y2, Y3, Y4 and Z3 three plots belong to A and one belongs to B.

Till now B hasn't got any plot in Third column and 2nd row.

So, Y3 belongs to B and Y2, Y4, Z3 belongs to A.

Let the number of trees in Y4 be $4x$ from 3, number of trees in Y3, Y2 will be $2x, x$ respectively.

The number of teak trees = $7x + 21$

∴ Number of mango trees = $14x + 42$

The table now looks like:

	1	2	3	4	
X	12 C				$14x+42$
Y	21 A	x A	2x B	4x A	$7x+21$
Z			9 A	28 B	

Each plot had trees in non-zero multiples of 3 or 4 and none of the plots had the same number of trees and from 2, B didn't have the largest number of trees in a plot $\Rightarrow x < 8$.

x can't be 7,5,3,2,1 as for these cases at least one of $x, 2x, 4x$ is neither multiple of 3 or 4.

x can be 6 or 4.

If $x=6$, number of Teak trees will be 63 and Mango trees will be 126 $\Rightarrow$ Number of Pine trees = $205-126-63=16$
but number of trees in $Z_3+Z_4>16$ so, $x \neq 6$.

If $x=4$, Number of Teak trees=49 and Mango trees=98 $\Rightarrow$ Number of Pine trees=58. Valid case.

Number of trees with $A = 30 + 5x = 50$.

From 1, number of trees with C, D= 30, 56 respectively.

So, number of trees in $Z_2 = 18$.

∴ Number of trees with B= $205-50-30-56=69$.

From 2, largest number of trees in a plot is 32. They can be in the plot of either B or D. If they are from B, they have to be from X2 but in that case number of trees in Z1=1 which is neither a multiple of 3 or 4.

So, highest number of trees in a plot are with D and it is 32 $\Rightarrow$ number of trees in X3, X4 are 32, 24 in any order.

So, number of trees in X2= $98-56-12=30$

∴ Number of trees in Z1= $69-30-28-8=3$.

The final table will look like:

	1	2	3	4	
X	12 C	30 B	32/24 D	24/32 D	98
Y	21 A	4 A	8 B	16 A	49
Z	3 B	18 C	9 A	28 B	58

Bina got 28 pine trees, Option B is correct answer.

 VIDEO SOLUTION

48. C

There are 12 plots and each of them got even number of plots. So, possible cases are 4,4,2,2 or 6,2,2,2.

From 4, A and B got more plots than D. So, the only possible case is A, B each got 4 and C,D each got 2.

From 6, D has to get two adjacent plots and From 8, plots of C, D are not adjacent to each other => D must have got plots in X3, X4.

C already has two plots in X1, Z2. So, the corner plot Z4 should belong to B.

From 7, B has a plot in each row and each column. So, X2 should belong to B.

Now, out of the remaining Y2, Y3, Y4 and Z3 three plots belong to A and one belongs to B.

Till now B hasn't got any plot in Third column and 2nd row.

So, Y3 belongs to B and Y2, Y4, Z3 belongs to A.

Let the number of trees in Y4 be $4x$ from 3, number of trees in Y3, Y2 will be $2x, x$ respectively.

The number of teak trees= $7x+21$

∴ Number of mango trees= $14x+42$

The table now looks like:

	1	2	3	4	
X	12 C				$14x+42$
Y	21 A	x A	2x B	4x A	$7x+21$
Z			9 A	28 B	

Each plot had trees in non-zero multiples of 3 or 4 and none of the plots had the same number of trees and from 2, B didn't have the largest number of trees in a plot $\Rightarrow x < 8$.

x can't be 7,5,3,2,1 as for these cases at least one of x,2x,4x is neither multiple of 3 or 4.

x can be 6 or 4.

If $x=6$, number of Teak trees will be 63 and Mango trees will be 126 $\Rightarrow$ Number of Pine trees = $205 - 126 - 63 = 16$ but number of trees in $Z_3 + Z_4 > 16$ so, $x \neq 6$.

If $x=4$, Number of Teak trees = 49 and Mango trees = 98 $\Rightarrow$ Number of Pine trees = 58. Valid case.

Number of trees with A = $30 + 5x = 50$.

From 1, number of trees with C, D = 30, 56 respectively.

So, number of trees in $Z_2 = 18$.

$\therefore$ Number of trees with B = $205 - 50 - 30 - 56 = 69$.

From 2, largest number of trees in a plot is 32. They can be in the plot of either B or D. If they are from B, they have to be from X_2 but in that case number of trees in $Z_1 = 1$ which is neither a multiple of 3 or 4.

So, highest number of trees in a plot are with D and it is 32 $\Rightarrow$ number of trees in X_3, X_4 are 32, 24 in any order.

So, number of trees in $X_2 = 98 - 56 - 12 = 30$

$\therefore$ Number of trees in $Z_1 = 69 - 30 - 28 - 8 = 3$.

The final table will look like:

	1	2	3	4	
X	12 C	30 B	32/24 D	24/32 D	98
Y	21 A	4 A	8 B	16 A	49
Z	3 B	18 C	9 A	28 B	58

$\therefore$ Number of Mango trees = 98.

49. **B**

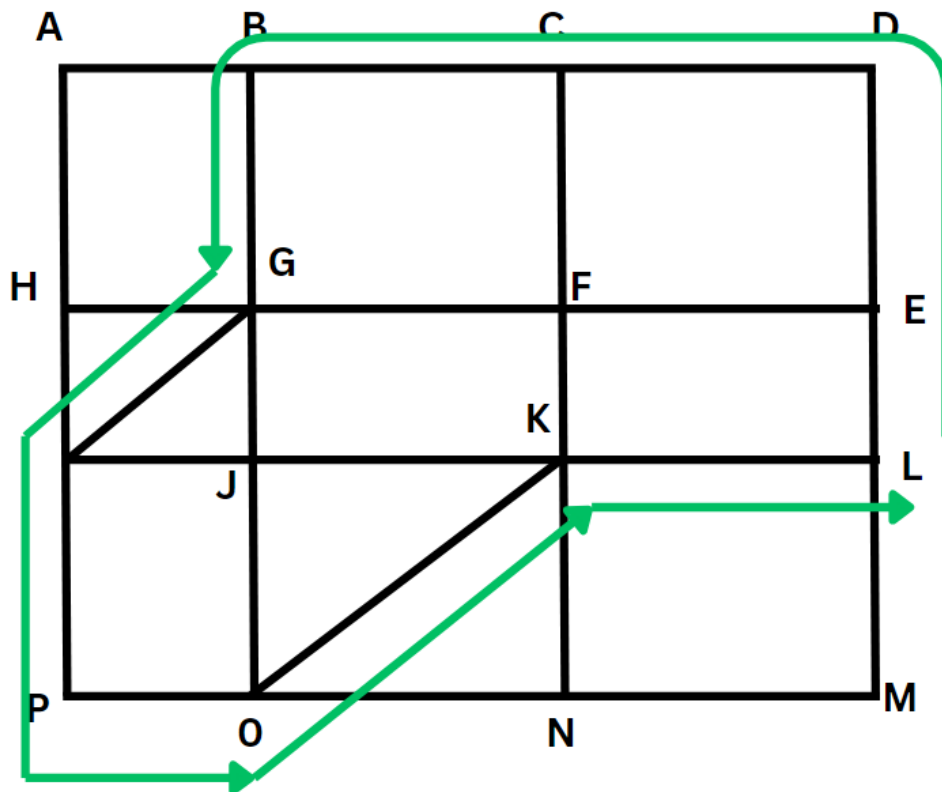
The first thing to realise here is the lengths of the paths.

KN should be equal to LM, giving the length of KO using Pythagoras theorem as 500m

Similarly, the length of $HJ=OP=150\text{m}$ and length of $GJ=EL=200\text{m}$, giving the length of HG as 250 m

The shortest path from L to B and then to P (or the other way around would involve) using these hypotenuses as much as possible instead of the two adjacent sides. The shortest can be visualised as shown below or multitude of others variations, as there are multiple ways that would make one travel the shortest distance)

The below figure is the simplest one for visualisation.



Other possible paths are L-E-F-C-.. and following the same path.

The shortest distance in each of these instance would be $LE+ED+DC+CB+BG+GI+IP+PO+OK+KL$

Which would be $200+400+300+300+400+250+400+150+500+300 = 3200$

Therefore, Option B is the correct answer.

50.5100

Counter to the first question, we should minimize the use of those hypotenuse walkways as they reduce the distance we travel.

The longest possible route would include then travelling through the edges of the rectangles and squares, and not touch any point more than once.

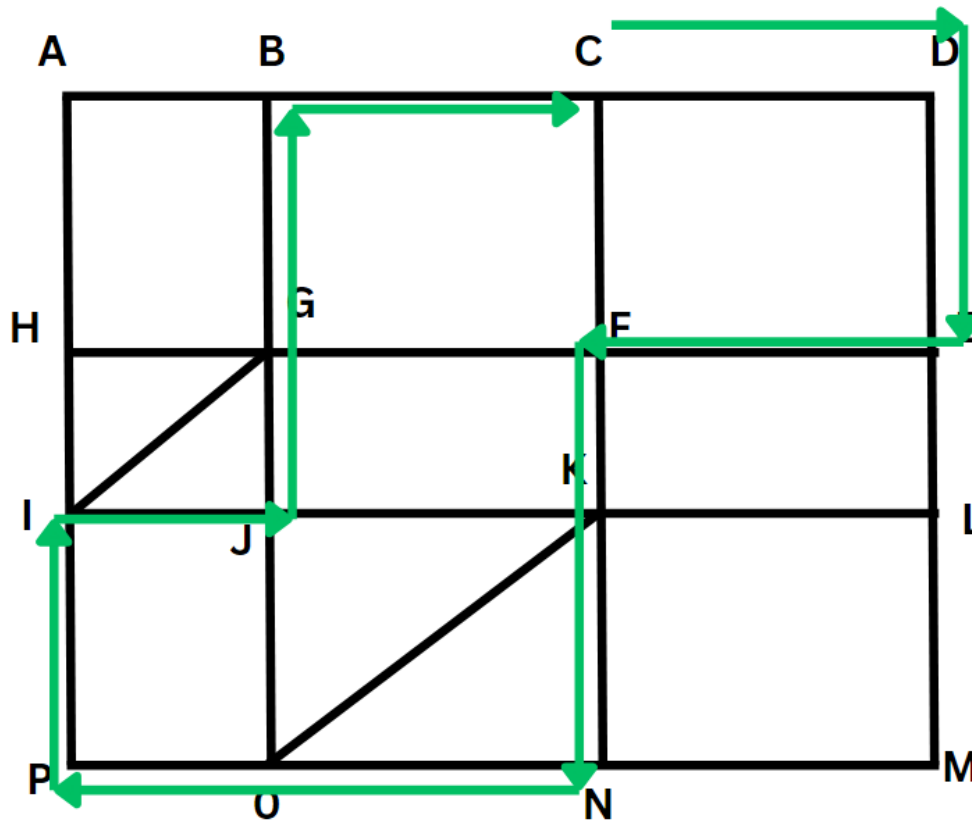
Best case scenario would be us touching each point exactly once.

This can be visualized as:

Core Cal

to avoid the hypot

For MBA/CAT C



The total path distance would be $300(CD) + 400(DE) + 300(EF) + 200(FK) + 400(KN) + 300(NO) + 150(OP) + 400(PI) + 150(IJ) + 200(JG) + 400(GB) + 300(BC)$

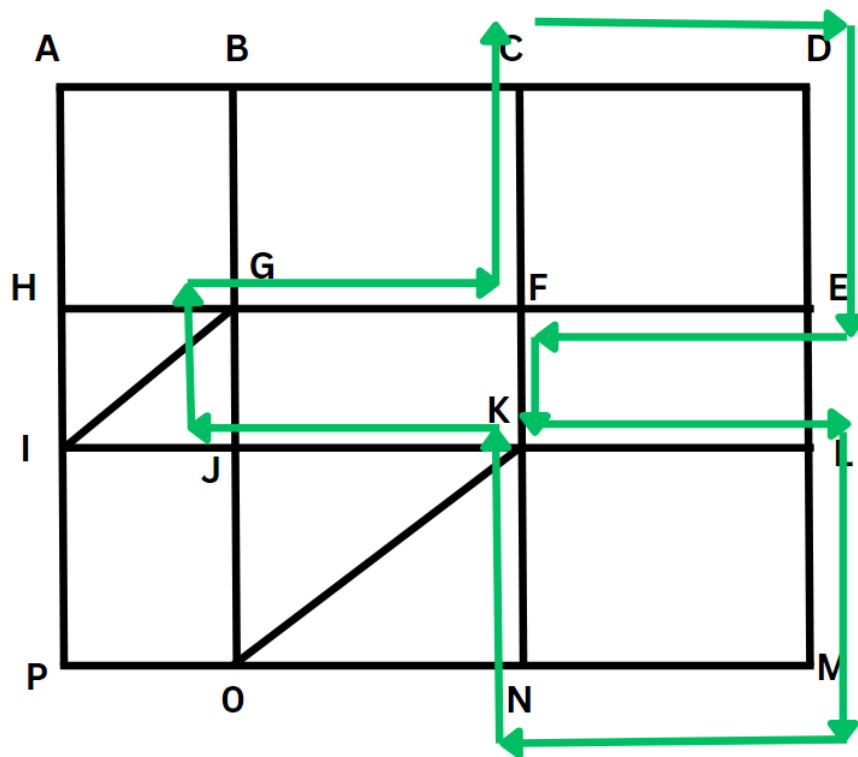
Adding up to 3,500 meters

Therefore, 3500 is the correct answer.

[VIDEO SOLUTION](#)

52. **C**

Since we can only walk along the side of lakes, that drastically reduces the paths we can take. The diagram below shows the path with the maximum distance travelled.



The path is CD-DE-EF-FK-KL-LM-MN-NK-KJ-JG-GF-FC

(the reverse of this path is also valid)

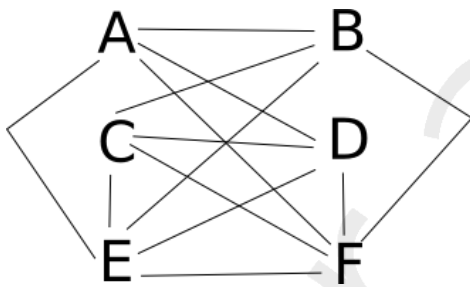
We can either manually add the lengths or use shorter methods to note that in the path we travel walkways of length 400 m 4 times (DE, LM, KN, EF), walkways of length 300m 6 times (CD, EF, KL, MN, KJ, GF) and walkways of length 200 m 2 times (FK, JG)

Giving the total length to be $1600 + 1800 + 400 = 3800$

Therefore, Option C is the correct answer.

[VIDEO SOLUTION](#)

53. A



Consider the following diagram: In this, the nodes E and F are connected to all the other nodes whereas the other four nodes are connected to four nodes each.

The total number of connections is 13.

[VIEW SOLUTION](#)

54. B

4 first level nodes are connected to A $\Rightarrow$ 4 ways.

Each of these first level nodes are connected to 1 second level node by four connections $\Rightarrow$ 4 ways

Each of these second level nodes are connected to 2 third level nodes $\Rightarrow$ 2 ways.

Each of these third level nodes are connected to B by 1 route $\Rightarrow$ 1 way

$\Rightarrow 4 * 4 * 2 * 1 = 32$ ways.

 [VIEW SOLUTION](#)